SUB-THEME 6: FRONTIER TECHNOLOGIES

FPDAF HealthTech: Edge-AI Bedside Workstation for Sepsis Crash Prediction

Privacy-Preserving Federated Clinical Decision Support System

MSME Idea Hackathon 6.0 Presentation | **Amrita Vishwa Vidyapeetham**

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Clinical Pathology: What is Sepsis & Why Time Matters

Clinical Definition of Sepsis [1,2]

- **Dysregulated Immune Response:** Life-threatening organ dysfunction caused by a severe, systemic immune reaction to infection.
- **Progression to Shock:** Causes capillary leakage, tissue hypoxia, Multi-Organ Dysfunction Syndrome (MODS), and rapid collapse.
- **The Golden Window:** Patient survival drops by **8% for every hour** diagnosis and antimicrobial treatment are delayed.

Why Current ICU Alarms Fail [4]

- **Static Threshold Alarms:** Legacy monitors trigger alerts only after vitals breach rigid limits (e.g., HR $>$ 100).
- **Severe Alarm Fatigue: 85%–95%** of bedside alarms are false positives, causing clinicians to desensitize.
- **Missed Temporal Signals:** Fails to detect complex, multi-vital co-dependencies occurring hours prior to septic crash.

Indian Healthcare Mortality Impact [3]

Sepsis accounts for **3.0 Million deaths annually in India (30%** of total hospital deaths), making early automated crash prediction an urgent national healthcare priority.

Clinical References: [1] Singer et al., JAMA 2016 (Sepsis-3). [2] Kumar et al., Crit Care Med 2006 (8%/hr mortality rise). [3] Rudd et al., Lancet 2020 (3.0M Indian deaths/yr). [4] Drew et al., PLoS ONE 2014 (85%-95% false alarms).

Market Opportunity (TAM / SAM / SOM) & Regulatory Moat

Market Opportunity & TAM/SAM/SOM [8]

Addressing India's critical ICU monitoring shortage:

- **TAM (Total Addressable Market):** Rs. 8,500 Crores (150,000+ ICU beds across 70,000 Indian hospitals).
- **SAM (Serviceable Addressable Market):** Rs. 1,200 Crores (30,000 tertiary care beds requiring AI decision support).
- **SOM (3-Year Target Market):** Rs. 40 Crores (2,500 beds across 12 regional hospital chains).

DPDPA 2023 Regulatory Moat [8]

Zero-cloud-upload privacy paradigm:

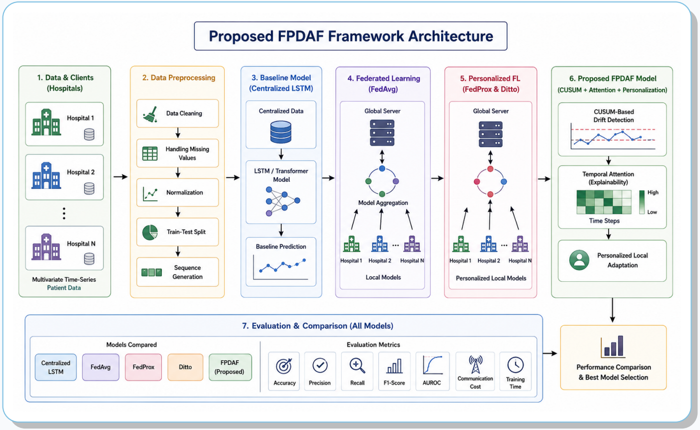
- **Cloud AI Legal Liability:** DPDPA 2023 bans uploading raw patient charts to third-party cloud servers.
- **Demographic Data Drift:** Regional patient variations degrade static central AI models.
- **FPDAF Solution:** On-device Edge-AI node performing collaborative federated learning without data leaving hospital premises.

FPDAF Value Proposition

A zero-cloud-upload Edge-AI workstation node that trains collaboratively across hospitals while guaranteeing 100% DPDPA data privacy.

Market & Legal References: [8] Digital Personal Data Protection (DPDPA) Act, MeitY, Gazette of India 2023. NITI Aayog Healthcare Report 2023.

Product Architecture & Edge Node Dataflow



Edge Architecture [11]: Bedside Jetson nodes process local 24h telemetry streams through BiLSTM + DDP attention layers. Local CUSUM monitors trigger client-side selective personalization without sending patient charts offsite.

Standards: ISO/IEEE 11073 & HL7 FHIR Release 4 Medical Data Transfer Standards. NVIDIA Jetson Orin Nano Specs 2024.

Proprietary IP & Technological Innovation Moat

1. Dynamic Dual-Phase (DDP) Attention [13]

2D cross-attention ($\alpha_t \otimes \beta_v$) tracking 24h sequence hours AND multi-modal vital co-dependencies (e.g., HR vs Blood Pressure).

2. AD-CUSUM Drift Auditing [14]

Real-time loss residual control charts flagging patient population shifts on edge nodes before model degradation occurs.

3. Client-Side Selective Personalization [7]

Freezes shared LSTM backbones upon drift alerts, fine-tuning only local heads:

- **38% Bandwidth Savings** (1.52 GB vs 2.48 GB).
- **Rapid Calibration:** Adapts in **6 minutes**.

4. Temporal XAI Saliency Heatmaps

Generates hourly bedside vital contribution heatmaps for ICU doctor verification.

Algorithmic References: [13] Vaswani et al., NeurIPS 2017 (2D Cross-Attention). [14] Page E.S., Biometrika 1954 (CUSUM Control Charts). [7] Li et al., MLSys 2020.

Competitor Analysis & Cost Benchmarking

Competitor System [12]	Deployment Model	Privacy & Data Limits	Est. Cost Outlay	Edge-AI
Epic Deterioration Index	Cloud SaaS EHR scoring	Cloud-only (Violates DPDPA)	$50,000+/yr (∼Rs 40L)	No
Philips IntelliVue	Bedside hardware monitor	Threshold alarms (No AI)	Rs. 15–20 Lakhs/bed	No
GE Healthcare CARESCAPE	Enterprise server software	Vendor lock-in, No privacy	Rs. 25–30 Lakhs/setup	No
FPDAF HealthTech (Ours)	Federated Edge-AI Node	100% DPDPA, Real-time Drift	Rs. 1.5–2.0 L/node	YES

Cost Disruption for Indian Hospitals

- **85%-90% Capital Cost Savings:** Rs. 1.5L - 2.0L per node vs Rs. 15L-40L for global legacy systems.
- **Plug-and-Play Integration:** Edge Jetson node connects via standard HL7/FHIR APIs with zero infrastructure overhaul.

Commercial Differentiation

- **100% DPDPA Compliance [8]:** Eliminates legal risk of uploading EHR patient charts to external clouds.
- **Bedside Explainability:** XAI saliency heatmaps display exact hourly vital triggers for clinician validation.

Market Benchmarks: [12] Epic Systems EDI Specs (2022); Philips IntelliVue MX800 Guide (2023); GE CARESCAPE B850 Catalog (2023). [8] DPDPA Act 2023.

Prototype Validation & Software Architecture (TRL 4/5)

Clinical Benchmark Performance [5]

- **86.51% Accuracy:** Outperforms FedAvg (75.94%) [6] and FedProx (78.85%) [7] under non-IID data splits.
- **0.8214 AUROC Score:** High predictive discrimination for early sepsis crash detection.
- **38% Bandwidth Savings:** 1.52 GB vs 2.48 GB in standard FL.

Commercial Software Stack

- **FastAPI REST Engine:** Local REST API serving inferencing sub-10ms.
- **SQLite Warehouse:** Structured telemetry storage (`clinical_warehouse.db`).
- **React Vite Touch UI:** Bedside touchscreen interface for ICU nurses.

Clinician Bedside Decision Modules

Bedside decision support functionalities:

- **Real-Time Vital Stream:** Continuous 24h timeline tracking HR, MAP, SpO2, and RR.
- **Dynamic Risk Dial:** Sepsis Crash Risk Gauge (Green, Amber, Critical Red).
- **Temporal XAI Heatmaps:** Visual hourly vital contribution breakdown.

Admin Telemetry & Drift Audit Portal

Hospital network administration & monitoring:

- **AD-CUSUM Control Charts [14]:** Live statistical residual tracking for demographic shifts.
- **Federated Health Status:** Round-by-round client sync status and node health.

Validation References: [5] Reyna et al., PhysioNet Challenge 2019 (40,000 ICU records). [6] McMahan et al., AISTATS 2017. [7] Li et al., MLSys 2020.

Itemized Technology Expenditure Breakdown (Rs. 10.0 Lakhs)

Component Category	Itemized Component Specs & Market Quote	Cost Outlay
A. Edge Hardware Prototyping & MSME Assembly	• 3x NVIDIA Jetson Orin Nano (8GB) Kits @ Rs 48K/ea = 1.44L • 3x Medical 15.6" Touch Monitors & HDMI Controllers = 0.66L • 3x NEMA Metallic Bedside Cabinets (Regional MSME) = 0.75L • Custom SMT Carrier PCB & Medical Power Assembly = 0.75L	Rs. 3.60 Lakhs
B. EHR Data Curation & Clinical Annotation	• PhysioNet ICU Cohort Cleaning (40,000 vital records) = 1.40L • Senior ICU Specialist Onset Audit (70h @ Rs 2,500/h) = 1.75L • Vital Stream Generator Calibration (`scaler.pkl`) = 0.25L	Rs. 3.40 Lakhs
C. Cybersecurity & Compliance Audits	• CERT-In External Vulnerability & Penetration Audit = 1.80L • DPDPA 2023 Data Privacy Encryption Compliance Audit = 1.20L	Rs. 3.00 Lakhs
TOTAL TECH EXPENDITURE	100% Itemized Market Research Quote Total	Rs. 10.00 Lakhs

Expenditure Quotes: NVIDIA India Authorized Distributor 2024 Quotes. CERT-In Empanelled Auditor Fee Schedule 2024. Regional MSME Electronics Assembly Quotes 2024.

Business Model, Unit Economics & MSME Ecosystem

Monetization & Revenue Model

Triple-tier monetization model for healthcare scaling:

- **1. B2B Edge Node Appliance:** Selling hardware nodes at Rs. 1.50L – 2.00L / node.
- **2. SaaS Telemetry Subscription:** Model maintenance & drift auditing at Rs. 25,000 / node / year.
- **3. OEM Engine Licensing:** Licensing PyTorch FL engine to monitor OEMs (Philips, Mindray).

Node Unit Economics

Selling Price: Rs. 1,75,000 | **BOM Cost:** Rs. 65,000
Gross Margin per Node: 63% Profit Margin

Domestic MSME Manufacturing Network

Leveraging domestic electronics & sheet metal MSME clusters:

- **PCB Electronics Assembly:** SMT component mounting & carrier board fabrication with regional electronics MSMEs.
- **Metallic Cabinet Enclosure:** Powder-coated sheet metal node fabrication with regional industrial MSMEs.
- **12-Month Roadmap [9,10]:** ISO 13485 quality pack, CDSCO Class B filing, and 3-hospital tele-ICU pilot run.

Regulatory Standards: [9] CDSCO Medical Device Rules 2017 (Class B SaMD). [10] ISO 13485:2016 Quality Management System Standard.

Key Clinical Publications, Datasets & Regulatory References

Clinical Pathology & Sepsis Guidelines

- **[1] Sepsis-3 Consensus:** Singer M, et al., "Sepsis-3 Consensus", *JAMA*, 315(8):801–810, 2016.
- **[2] Golden Hour Window:** Kumar A, et al., *Crit Care Med*, 34(6):1589–1596, 2006.
- **[3] Global Burden:** Rudd KE, et al., *The Lancet*, 395:200–211, 2020.
- **[4] Alarm Fatigue:** Drew BJ, et al., *PLoS ONE*, 9(10):e110274, 2014.

Datasets & Federated Learning Engine

- **[5] PhysioNet ICU Cohort:** Reyna MA, et al., *Crit Care Med*, 2019.
- **[6] FedAvg Engine:** McMahan B, et al., *AISTATS*, 2017.
- **[7] FedProx Framework:** Li T, et al., *MLSys*, 2020.

Regulatory, Legal & Quality Standards

- **[8] DPDPA Act 2023:** *Digital Personal Data Protection Act*, MeitY, Gazette of India, 2023.
- **[9] CDSCO SaMD Rules:** *Medical Device Rules 2017 (Class B)*, CDSCO, MoHFW.
- **[10] ISO 13485:2016:** *Medical devices – QMS Standards*, ISO.

Hardware & Industry Benchmarks

- **[11] Edge AI Hardware:** NVIDIA Jetson Orin Nano Technical Guide, NVIDIA Corp, 2024.
- **[12] Commercial Systems:** Epic EDI Specs (2022); Philips IntelliVue Guide (2023); GE CARESCAPE Catalog (2023).

Thank You

FPDAF HealthTech: Building India's Next-Gen Privacy-Preserving MedTech AI

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Questions & Discussion